Chained NPD for Memory MAC

The Math on the Board

1 The Chained Decoding Idea

Two users U and V transmit through a shared channel. At the **corner rate** (Class C path), we decode in two stages:

Stage 1: Decode U (treat V as noise)

$$\hat{u}_i = \arg \max_{u_i \in \{0,1\}} P(u_i \mid z_1^N, \hat{u}_1^{i-1})$$

V is unknown $\Rightarrow$ marginalized as uniform noise. This is a **single-user** decoding problem on the effective channel $P(Z \mid X) = \sum_Y P(Y) \cdot P(Z \mid X, Y)$.

Stage 2: Decode V given $\hat{U}$

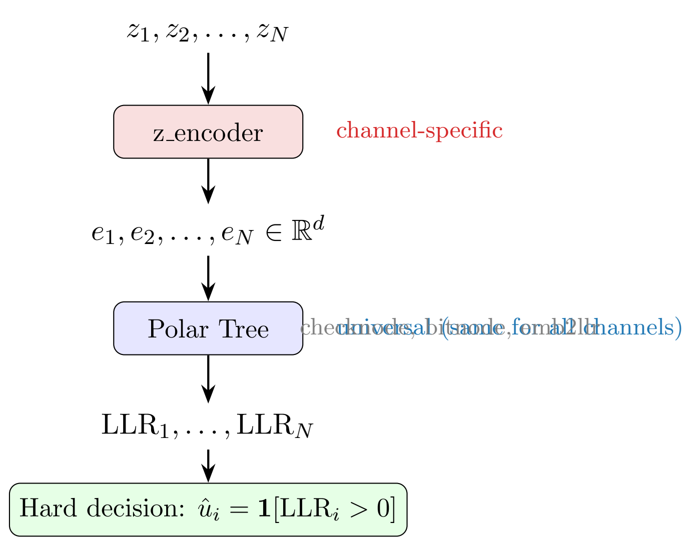
$$\hat{v}_i = \arg \max_{v_i \in \{0,1\}} P(v_i \mid z_1^N, \hat{u}_1^N, \hat{v}_1^{i-1})$$

$\hat{U}$ is known from Stage 1 $\Rightarrow$ this is also a single-user problem, on the effective channel $P(Z \mid Y, \hat{X})$.

Each stage produces a **binary LLR** at each position:

$$\mathrm{LLR}_i = \log \frac{P(\mathrm{bit}_i = 0 \mid \text{observations})}{P(\mathrm{bit}_i = 1 \mid \text{observations})}$$

2 Architecture: What Changes Per Channel



Key insight: Only the z_encoder changes between channels. The tree operations are **identical**.

3 The Three Channels

3.1 GMAC (Memoryless)

$$\boxed{Z_i = (1 - 2X_i) + (1 - 2Y_i) + W_i, \quad W_i \sim \mathcal{N}(0, \sigma^2)}$$

- Each Z_i depends only on (X_i, Y_i) — no memory.
- **z_encoder:** pointwise MLP. $e_i = \text{MLP}(z_i)$. Each position processed independently.
- **Stage 1 effective channel:** $P(Z_i \mid X_i) = \frac{1}{2}\big[\mathcal{N}(Z_i;\, s_{X_i}{+}1,\, \sigma^2) + \mathcal{N}(Z_i;\, s_{X_i}{-}1,\, \sigma^2)\big]$ where $s_{X_i} = 1 - 2X_i$. This is a Gaussian **mixture** (marginalize Y_i uniform).

3.2 ISI-MAC (Finite-State Memory)

$$\boxed{Z_i = (1{-}2X_i) + (1{-}2Y_i) + h\big((1{-}2X_{i-1}) + (1{-}2Y_{i-1})\big) + W_i}$$

- Z_i depends on **current and previous** inputs. Parameter $h = 0.3$.
- State: $S_i = (X_{i-1}, Y_{i-1}) \in \{0,1\}^2$ — 4 states.
- **z_encoder:** Bidirectional GRU over the full sequence.

$$(e_1, \ldots, e_N) = \text{BiGRU}(z_1, \ldots, z_N)$$

  The BiGRU sees neighboring z values and learns to cancel the ISI interference.
- **Stage 1 effective channel:** $P(Z_i \mid X_i, S_i) = \frac{1}{2}\sum_{Y_i} P(Z_i \mid X_i, Y_i, S_i)$
  This is a finite-state channel $\Rightarrow$ trellis SC decoder exists (our analytical baseline).

3.3 MA-AGN MAC (Continuous-State Memory)

$$\boxed{Z_i = (1{-}2X_i) + (1{-}2Y_i) + N_i, \quad N_i = \alpha \cdot N_{i-1} + W_i}$$

- Noise N_i is an AR(1) process with parameter α. Noise samples are **correlated**.
- State: $N_{i-1} \in \mathbb{R}$ — **continuous**, not finite.
- **z_encoder:** Same BiGRU. Learns to exploit the noise correlation.
- **Stage 1 effective channel:** No closed-form trellis (continuous state).

$$P(Z_i \mid X_i, N_{i-1}) = \frac{1}{2}\sum_{Y_i} \mathcal{N}\big(Z_i;\, s_{X_i} + s_{Y_i} + \alpha N_{i-1},\, \sigma^2(1-\alpha^2)\big)$$

  No analytical decoder exists $\Rightarrow$ neural decoder is the only option.

4 Training: fast_ce (Parallel)

Each stage is trained with **fast_ce**: teacher-forced binary cross-entropy at every tree depth simultaneously.

1. Generate codewords: $u \to x = \text{polar\_encode}(u)$, same for $v \to y$.
2. Channel: $z = \text{channel}(x, y)$.
3. z_encoder: $e = \text{z\_enc}(z)$ (pointwise or BiGRU).
4. At each tree depth $\ell = 0, 1, \ldots, \log_2 N$:
   - Apply checknode/bitnode using **true codeword bits** (teacher forcing).
   - Compute $\text{LLR} = \text{emb2llr}(e)$ at all positions.
   - Loss: $\text{BCE}(\text{LLR}, \text{true bits})$.
5. Total loss = average over all depths. One backward pass.

Gradient depth: $O(\log N)$ instead of $O(N \log N)$ for sequential training.

5 Summary Table

	GMAC	**ISI-MAC**	**MA-AGN**
Memory	None	Finite ($\lvert S\rvert = 4$)	Continuous ($S \in \mathbb{R}$)
z_encoder	Pointwise MLP	BiGRU	BiGRU
Analytical SC	Yes (exact)	Yes (trellis)	**No**
Tree ops	Same MLPs for all channels		
Training	fast_ce, $O(\log N)$ gradient depth		
NPD beats SC?	$N \leq 64$	$N \leq 64$	$N \leq 64$ (α=0.9)
Wall at	$N = 128^{\dagger}$	$N = 512$	$N = 128$

†GMAC C wall at $N = 128$ is due to missing 3-phase frozen-set design, not decoder limitation.

6 Worked Example: GMAC Chained SC at $N = 2$

Setup: $u = (0, 1)$, $v = (1, 0)$. Encoding: $x = (1, 1)$, $y = (1, 0)$.
Received: $z_1 = -1.75$, $z_2 = -0.15$. SNR = 6 dB ($\sigma^2 = 0.251$).

Stage 1: Decode U (marginalize V as uniform)

Effective single-user channel: $P(z_i \mid x_i) = \frac{1}{2}[P(z_i \mid x_i, y{=}0) + P(z_i \mid x_i, y{=}1)]$:

$$P(z_1 \mid x{=}0) = 0.0009, \quad P(z_1 \mid x{=}1) = 0.3524$$

$$P(z_2 \mid x{=}0) = 0.3807, \quad P(z_2 \mid x{=}1) = 0.3811$$

Effective LLRs:

$$\mathrm{LLR}_{z_1} = \log \frac{0.0009}{0.3524} = -5.98, \qquad \mathrm{LLR}_{z_2} = \log \frac{0.3807}{0.3811} = -0.001$$

z_1 strongly says $x_1 = 1$. z_2 is ambiguous (the (0,1)/(1,0) collision at $z \approx 0$).

CalcLeft (f-function, decode u_1):

$$\mathrm{LLR}_{u_1} = 2 \tanh^{-1}\big(\tanh(\tfrac{-5.98}{2}) \cdot \tanh(\tfrac{-0.001}{2})\big) = +0.001$$

$\hat{u}_1 = 0$ (barely, LLR ≈ 0). **Correct** ($u_1 = 0$).

CalcRight (g-function, decode u_2):

$$\mathrm{LLR}_{u_2} = \mathrm{LLR}_{z_2} + (1 - 2\hat{u}_1) \cdot \mathrm{LLR}_{z_1} = -0.001 + 1 \times (-5.98) = -5.98$$

$\hat{u}_2 = 1$ (strong). **Correct** ($u_2 = 1$).

Stage 2: Decode V given $\hat{U} = (0, 1)$

Now X is known: $\hat{x}_1 = \hat{u}_1 \oplus \hat{u}_2 = 1$, $\hat{x}_2 = \hat{u}_2 = 1$. BPSK: $s_{\hat{x}} = (-1, -1)$.
Clean single-user channel: $P(z_i \mid y_i, \hat{x}_i) = \mathcal{N}(z_i;\, s_{\hat{x}_i} + s_{y_i},\, \sigma^2)$:

$$\mathrm{LLR}^{(v)}_{z_1} = \log \frac{\mathcal{N}(-1.75;\, 0,\, \sigma^2)}{\mathcal{N}(-1.75;\, -2,\, \sigma^2)} = \log \frac{0.0018}{0.7031} = -5.98$$

$$\mathrm{LLR}^{(v)}_{z_2} = \log \frac{\mathcal{N}(-0.15;\, 0,\, \sigma^2)}{\mathcal{N}(-0.15;\, -2,\, \sigma^2)} = \log \frac{0.7614}{0.0009} = +6.77$$

CalcLeft: $\mathrm{LLR}_{v_1} = 2 \tanh^{-1}\big(\tanh(\frac{-5.98}{2}) \cdot \tanh(\frac{6.77}{2})\big) = -5.60$. $\hat{v}_1 = 1$. **Correct.**
CalcRight: $\mathrm{LLR}_{v_2} = 6.77 + (1 - 2 \cdot 1) \cdot (-5.98) = +12.75$. $\hat{v}_2 = 0$. **Correct.**

Key Observations

[nosep]**Stage 1 is harder:** z_2 gives LLR ≈ 0 because $P(z|x{=}0)$ and $P(z|x{=}1)$ are nearly equal after marginalizing V. The Gaussian mixture from V-marginalization creates an information "dead zone" near $z = 0$. **Stage 2 is easier:** once X is known, each z_i gives a clean Gaussian LLR (no mixture). Stage 2 LLRs (± 5.98, ± 6.77) are much stronger than Stage 1. **Chaining works:** even though Stage 1 barely resolves u_1, the cascading through CalcRight recovers u_2 strongly, and Stage 2 decodes V cleanly.